Classification

Week 4: K-Nearest Neighbours, Support Vector Machines

In this week, let us work on standard and a lengthy dataset on two classification algorithms KNN and SVM. After generation of the dataset related to Social Network Advertisements, format the general methods before deployment namely Importing the libraries, Importing the dataset, Splitting the Training Set and Test Set, Feature Scaling and Display the Outputs in both Test and Trained Data.

Later, Train the concern KNN and SVM Models on the Training Set and Predict the respective new results and also Confusion matrix is generated by visualizing the Training Set Results and Test Set results.

Dataset:

Social_Network_Ads.csv

Age	EstimatedSalary	Purchased
19	19000	0
35	20000	0
26	43000	0
27	57000	0
19	76000	0
27	58000	0
27	84000	0
32	150000	1
25	33000	0
35	65000	0
26	80000	0
26	52000	0
20	86000	0
32	18000	0
18	82000	0
29	80000	0
47	25000	1
45	26000	1
46	28000	1
48	29000	1
45	22000	1
47	49000	1
48	41000	1
45	22000	1
46	23000	1
47	20000	1
49	28000	1
47	30000	1

29	43000	0
31	18000	0
31	74000	0
27	137000	1
21	16000	0
28	44000	0
27	90000	0
35	27000	0
33	28000	0
30	49000	0
26	72000	0
27	31000	0
27	17000	0
33	51000	0
35	108000	0
30	15000	0
28	84000	0
23	20000	0
25	79000	0
27	54000	0
30	135000	1
31	89000	0
24	32000	0
18	44000	0
29	83000	0
35	23000	0
27	58000	0
24	55000	0
23	48000	0
28	79000	0
22	18000	0
32	117000	0
27	20000	0
25	87000	0
23	66000	0
32	120000	1
59	83000	0
24	58000	0
24	19000	0
23	82000	0
22	63000	0
31	68000	0
25	80000	0
24	27000	0
20	23000	0

33	113000	0
32	18000	0
34	112000	1
18	52000	0
22	27000	0
28	87000	0
26	17000	0
30	80000	0
39	42000	0
20	49000	0
35	88000	0
30	62000	0
31	118000	1
24	55000	0
28	85000	0
26	81000	0
35	50000	0
22	81000	0
30	116000	0
26	15000	0
29	28000	0
29	83000	0
35	44000	0
35	25000	0
28	123000	1
35	73000	0
28	37000	0
27	88000	0
28	59000	0
32	86000	0
33	149000	1
19	21000	0
21	72000	0
26	35000	0
27	89000	0
26	86000	0
38	80000	0
39	71000	0
37	71000	0
38	61000	0
37	55000	0
42	80000	0
40	57000	0
35	75000	0
36	52000	0

40	59000	0
41	59000	0
36	75000	0
37	72000	0
40	75000	0
35	53000	0
41	51000	0
39	61000	0
42	65000	0
26	32000	0
30	17000	0
26	84000	0
31	58000	0
33	31000	0
30	87000	0
21	68000	0
28	55000	0
23	63000	0
20	82000	0
30	107000	1
28	59000	0
19	25000	0
19	85000	0
18	68000	0
35	59000	0
30	89000	0
34	25000	0
24	89000	0
27	96000	1
41	30000	0
29	61000	0
20	74000	0
26	15000	0
41	45000	0
31	76000	0
36	50000	0
40	47000	0
31	15000	0
46	59000	0
29	75000	0
26	30000	0
32	135000	1
32	100000	1
25	90000	0
37	33000	0

35	38000	0
33	69000	0
18	86000	0
22	55000	0
35	71000	0
29	148000	1
29	47000	0
21	88000	0
34	115000	0
26	118000	0
34	43000	0
34	72000	0
23	28000	0
35	47000	0
25	22000	0
24	23000	0
31	34000	0
26	16000	0
31	71000	0
32	117000	1
33	43000	0
33	60000	0
31	66000	0
20	82000	0
33	41000	0
35	72000	0
28	32000	0
24	84000	0
19	26000	0
29	43000	0
19	70000	0
28	89000	0
34	43000	0
30	79000	0
20	36000	0
26	80000	0
35	22000	0
35	39000	0
49	74000	0
39	134000	1
41	71000	0
58	101000	1
47	47000	0
55	130000	1
52	114000	0

40	142000	1
46	22000	0
48	96000	1
52	150000	1
59	42000	0
35	58000	0
47	43000	0
60	108000	1
49	65000	0
40	78000	0
46	96000	0
59	143000	1
41	80000	0
35	91000	1
37	144000	1
60	102000	1
35	60000	0
37	53000	0
36	126000	1
56	133000	1
40	72000	0
42	80000	1
35	147000	1
39	42000	0
40	107000	1
49	86000	1
38	112000	0
46	79000	1
40	57000	0
37	80000	0
46	82000	0
53	143000	1
42	149000	1
38	59000	0
50	88000	1
56	104000	1
41	72000	0
51	146000	1
35	50000	0
57	122000	1
41	52000	0
35	97000	1
44	39000	0
37	52000	0
48	134000	1

37	146000	1
50	44000	0
52	90000	1
41	72000	0
40	57000	0
58	95000	1
45	131000	1
35	77000	0
36	144000	1
55	125000	1
35	72000	0
48	90000	1
42	108000	1
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37	74000	0
47	144000	1
40	61000	0
43	133000	0
59	76000	1
60	42000	1
39	106000	1
57	26000	1
57	74000	1
38	71000	0
49	88000	1
52	38000	1
50	36000	1
59	88000	1
35	61000	0
37	70000	1
52	21000	1
48	141000	0
37	93000	1
37	62000	0
48	138000	1
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37	78000	1
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35	57000	0
36	63000	0
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37	79000	1
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42	54000	0
51	134000	0
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39	96000	1
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49	141000	1
39	79000	0
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35	55000	0
45	32000	1
36	60000	0
52	138000	1
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48	30000	1
48	131000	1
41	60000	0
41	72000	0
42	75000	0
36	118000	1
47	107000	1
38	51000	0
48	119000	1
42	65000	0
40	65000	0
57	60000	1
36	54000	0
58	144000	1
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38	55000	0
39	122000	1
53	104000	1
35	75000	0
38	65000	0

47	51000	1
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38	113000	1
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37	57000	0
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40	71000	1
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43	129000	1
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49	28000	1
57	33000	1
56	60000	1
49	39000	1
39	71000	0

47	34000	1
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48	33000	1
47	23000	1
45	45000	1
60	42000	1
39	59000	0
46	41000	1
51	23000	1
50	20000	1
36	33000	0
49	36000	1

K-Nearest Neighbors (K-NN)

Importing the libraries

```
import numpy as np

import matplotlib.pyplot as plt

import pandas as pd
```

Importing the dataset

```
dataset = pd.read_csv('Social_Network_Ads.csv')

X = dataset.iloc[:, :-1].values

y = dataset.iloc[:, -1].values
```

Splitting the dataset into the Training set and Test set

```
from sklearn.model_selection import train_test_split

X_train, X_test, y_train, y_test = train_test_split(X, y, test_size = 0.25, random_state = 0)
```

print(X_train)

```
[[    44  39000]
 [    32 120000]
 [    38  50000]
 [    32 135000]
 [    52  21000]
 [    53 104000]
 [    39  42000]
 [    38  61000]
 [    36  50000]
 [    36  63000]
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[    35  25000]
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[    35  23000]
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 [    42  54000]
 [    34  43000]
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 [    29  43000]
 [    36  52000]
 [    27  54000]
 [    26 118000]]
```

print(y_train)

```
[0 1 0 1 1 1 0 0 0 0 0 0 1 1 1 0 1 0 0 1 0 1 0 1 0 0 1 1 1 1 0 1 0 1 0 0 1
 0 0 1 0 0 0 0 0 1 1 1 1 0 0 0 1 0 1 0 1 0 0 1 0 0 0 1 0 0 0 1 1 0 0 1 0 1
 1 1 0 0 1 1 0 0 1 1 0 1 0 0 1 1 0 1 1 1 0 0 0 0 0 1 0 0 1 1 1 1 1 0 1 1 0
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 0 0 1 0 1 1 0 0 0 0 0 1 0 1 0 0 1 0 0 1 0 1 0 0 0 0 0 0 1 1 1 1 0 0 0 0 1
 0 0 0 0]
```

print(X_test)

```
[[    30  87000]
 [    38  50000]
 [    35  75000]
 [    30  79000]
 [    35  50000]
 [    27  20000]
 [    31  15000]
 [    36 144000]
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 [    35  22000]
 [    35  47000]
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 [    47 105000]
 [    23  28000]
 [    49 141000]
```

```
[    28  87000]
[    29  80000]
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[    32  86000]
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 [    42  79000]
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 [    48  90000]
 [    42 104000]]
```

print(y_test)

```
[0 0 0 0 0 0 0 1 0 0 0 0 0 0 0 0 0 0 1 0 0 1 0 1 0 1 0 0 0 0 0 1 1 0 0 0 0
 0 0 1 0 0 0 0 1 0 0 1 0 1 1 0 0 0 1 1 0 0 1 0 0 1 0 1 0 1 0 0 0 0 1 0 0 1
 0 0 0 0 1 1 1 0 0 0 1 1 0 1 1 0 0 1 0 0 0 1 0 1 1 1]
```

Feature Scaling

```
from sklearn.preprocessing import StandardScaler
sc = StandardScaler()
X_train = sc.fit_transform(X_train)
X_test = sc.transform(X_test)
```

```
[[ 0.58164944 -0.88670699]
 [-0.60673761  1.46173768]
 [-0.01254409 -0.5677824 ]
 [-0.60673761  1.89663484]
 [ 1.37390747 -1.40858358]
 [ 1.47293972  0.99784738]
 [ 0.08648817 -0.79972756]
 [-0.01254409 -0.24885782]
 [-0.21060859 -0.5677824 ]
 [-0.21060859 -0.19087153]
 [-0.30964085 -1.29261101]
```

```
[-0.30964085 -0.5677824 ]
[ 0.38358493  0.09905991]
[ 0.8787462  -0.59677555]
[ 2.06713324 -1.17663843]
[ 1.07681071 -0.13288524]
[ 0.68068169  1.78066227]
[-0.70576986  0.56295021]
[ 0.77971394  0.35999821]
[ 0.8787462  -0.53878926]
[-1.20093113 -1.58254245]
[ 2.1661655   0.93986109]
[-0.01254409  1.22979253]
[ 0.18552042  1.08482681]
[ 0.38358493 -0.48080297]
[-0.30964085 -0.30684411]
[ 0.97777845 -0.8287207 ]
[ 0.97777845  1.8676417 ]
[-0.01254409  1.25878567]
[-0.90383437  2.27354572]
[-1.20093113 -1.58254245]
[ 2.1661655  -0.79972756]
[-1.39899564 -1.46656987]
[ 0.38358493  2.30253886]
[ 0.77971394  0.76590222]
[-1.00286662 -0.30684411]
[ 0.08648817  0.76590222]
[-1.00286662  0.56295021]
[ 0.28455268  0.07006676]
[ 0.68068169 -1.26361786]
[-0.50770535 -0.01691267]
[-1.79512465  0.35999821]
[-0.70576986  0.12805305]
[ 0.38358493  0.30201192]
[-0.30964085  0.07006676]
[-0.50770535  2.30253886]
[ 0.18552042  0.04107362]
[ 1.27487521  2.21555943]
[ 0.77971394  0.27301877]
[-0.30964085  0.1570462 ]
[-0.01254409 -0.53878926]
[-0.21060859  0.1570462 ]
[-0.11157634  0.24402563]
[-0.01254409 -0.24885782]
[ 2.1661655   1.11381995]
[-1.79512465  0.35999821]
[ 1.86906873  0.12805305]
[ 0.38358493 -0.13288524]
[-1.20093113  0.30201192]
[ 0.77971394  1.37475825]
[-0.30964085 -0.24885782]
```

```
[-1.6960924  -0.04590581]
[-1.00286662 -0.74174127]
[ 0.28455268  0.50496393]
[-0.11157634 -1.06066585]
[-1.10189888  0.59194336]
[ 0.08648817 -0.79972756]
[-1.00286662  1.54871711]
[-0.70576986  1.40375139]
[-1.29996338  0.50496393]
[-0.30964085  0.04107362]
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[-1.29996338  0.27301877]
[ 1.37390747  1.98361427]
[ 1.27487521 -1.3505973 ]
[-0.30964085 -0.27785096]
[-0.50770535  1.25878567]
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[ 0.97777845 -1.06066585]
[ 0.28455268  0.30201192]
[ 0.97777845  0.76590222]
[-0.70576986 -1.49556302]
[-0.70576986  0.04107362]
[ 0.48261718  1.72267598]
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[ 0.8787462  -1.14764529]
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[ 0.18552042 -0.3648304 ]
[ 2.06713324  2.12857999]
[ 1.86906873 -1.26361786]
[ 1.37390747 -0.91570013]
[ 0.8787462   1.25878567]
[ 1.47293972  2.12857999]
[-0.30964085 -1.23462472]
[ 1.96810099  0.91086794]
[ 0.68068169 -0.71274813]
[-1.49802789  0.35999821]
```

```
[ 0.77971394 -1.3505973 ]
[ 0.38358493 -0.13288524]
[-1.00286662  0.41798449]
[-0.01254409 -0.30684411]
[-1.20093113  0.41798449]
[-0.90383437 -1.20563157]
[-0.11157634  0.04107362]
[-1.59706014 -0.42281668]
[ 0.97777845 -1.00267957]
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[-0.01254409 -0.13288524]
[-1.10189888 -1.52455616]
[ 0.77971394 -1.20563157]
[ 0.97777845  2.07059371]
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[ 0.77971394  0.53395707]
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[-1.00286662 -1.089659  ]
[ 1.17584296 -1.43757673]
[ 0.18552042 -0.30684411]
[ 1.17584296 -0.74174127]
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[ 0.77971394 -1.089659  ]
[ 0.08648817  0.04107362]
[-1.79512465  0.12805305]
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[ 0.8787462  -1.29261101]
[ 0.18552042 -0.24885782]
[-0.4086731   1.22979253]
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[ 0.38358493  0.1570462 ]
[ 0.8787462  -0.65476184]
[ 0.08648817  0.1570462 ]
[-1.89415691 -1.29261101]
[-0.11157634  0.30201192]
[-0.21060859 -0.27785096]
[ 0.28455268 -0.50979612]
```

```
[-0.21060859  1.6067034 ]
[ 0.97777845 -1.17663843]
[-0.21060859  1.63569655]
[ 1.27487521  1.8676417 ]
[-1.10189888 -0.3648304 ]
[-0.01254409  0.04107362]
[ 0.08648817 -0.24885782]
[-1.59706014 -1.23462472]
[-0.50770535 -0.27785096]
[ 0.97777845  0.12805305]
[ 1.96810099 -1.3505973 ]
[ 1.47293972  0.07006676]
[-0.60673761  1.37475825]
[ 1.57197197  0.01208048]
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[ 1.96810099  0.73690908]
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[ 0.18552042  0.1570462 ]
[-0.50770535 -1.20563157]
[ 0.58164944  2.01260742]
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[-0.50770535 -0.53878926]
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[ 1.57197197  0.99784738]
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[-0.30964085 -0.48080297]
[-0.11157634  2.15757314]
[-1.49802789 -0.1038921 ]
[-0.11157634  1.95462113]
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[-0.50770535 -0.8287207 ]
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[-0.80480212 -1.58254245]
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[ 1.07681071  0.12805305]
[ 0.08648817  1.51972397]
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[ 0.08648817  0.04107362]
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[ 1.57197197 -1.26361786]
[-0.30964085 -0.74174127]
[-0.11157634  0.1570462 ]
[-0.90383437 -0.65476184]
```

```
[-0.70576986 -0.04590581]
[ 0.38358493 -0.45180983]
[-0.80480212  1.89663484]
[ 1.37390747  1.28777882]
[ 1.17584296 -0.97368642]
[ 1.77003648  1.83864855]
[-0.90383437 -0.24885782]
[-0.80480212  0.56295021]
[-1.20093113 -1.5535493 ]
[-0.50770535 -1.11865214]
[ 0.28455268  0.07006676]
[-0.21060859 -1.06066585]
[ 1.67100423  1.6067034 ]
[ 0.97777845  1.78066227]
[ 0.28455268  0.04107362]
[-0.80480212 -0.21986468]
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[ 0.28455268 -0.19087153]
[ 1.96810099 -0.65476184]
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[-0.11157634  0.12805305]
[ 0.28455268 -0.30684411]
[ 1.07681071  0.56295021]
[-1.00286662  0.27301877]
[ 1.47293972  0.35999821]
[ 0.18552042 -0.3648304 ]
[ 2.1661655  -1.03167271]
[-0.30964085  1.11381995]
[-1.6960924   0.07006676]
[-0.01254409  0.04107362]
[ 0.08648817  1.05583366]
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[ 1.57197197  1.11381995]
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[ 0.08648817  1.8676417 ]
[-0.90383437 -0.77073441]
[-0.50770535 -0.77073441]
[-0.30964085 -0.91570013]
[ 0.28455268 -0.71274813]
[ 0.28455268  0.07006676]
[ 0.08648817  1.8676417 ]
[-1.10189888  1.95462113]
[-1.6960924  -1.5535493 ]
[-1.20093113 -1.089659  ]
[-0.70576986 -0.1038921 ]
[ 0.08648817  0.09905991]
[ 0.28455268  0.27301877]
```

```
 [ 0.8787462  -0.5677824 ]
 [ 0.28455268 -1.14764529]
 [-0.11157634  0.67892279]
 [ 2.1661655  -0.68375498]
 [-1.29996338 -1.37959044]
 [-1.00286662 -0.94469328]
 [-0.01254409 -0.42281668]
 [-0.21060859 -0.45180983]
 [-1.79512465 -0.97368642]
 [ 1.77003648  0.99784738]
 [ 0.18552042 -0.3648304 ]
 [ 0.38358493  1.11381995]
 [-1.79512465 -1.3505973 ]
 [ 0.18552042 -0.13288524]
 [ 0.8787462  -1.43757673]
 [-1.99318916  0.47597078]
 [-0.30964085  0.27301877]
 [ 1.86906873 -1.06066585]
 [-0.4086731   0.07006676]
 [ 1.07681071 -0.88670699]
 [-1.10189888 -1.11865214]
 [-1.89415691  0.01208048]
 [ 0.08648817  0.27301877]
 [-1.20093113  0.33100506]
 [-1.29996338  0.30201192]
 [-1.00286662  0.44697764]
 [ 1.67100423 -0.88670699]
 [ 1.17584296  0.53395707]
 [ 1.07681071  0.53395707]
 [ 1.37390747  2.331532  ]
 [-0.30964085 -0.13288524]
 [ 0.38358493 -0.45180983]
 [-0.4086731  -0.77073441]
 [-0.11157634 -0.50979612]
 [ 0.97777845 -1.14764529]
 [-0.90383437 -0.77073441]
 [-0.21060859 -0.50979612]
 [-1.10189888 -0.45180983]
 [-1.20093113  1.40375139]]
```

print(X_test)

```
[[-0.80480212  0.50496393]
 [-0.01254409 -0.5677824 ]
 [-0.30964085  0.1570462 ]
 [-0.80480212  0.27301877]
 [-0.30964085 -0.5677824 ]
 [-1.10189888 -1.43757673]
 [-0.70576986 -1.58254245]
 [-0.21060859  2.15757314]
```

```
[-1.99318916 -0.04590581]
[ 0.8787462  -0.77073441]
[-0.80480212 -0.59677555]
[-1.00286662 -0.42281668]
[-0.11157634 -0.42281668]
[ 0.08648817  0.21503249]
[-1.79512465  0.47597078]
[-0.60673761  1.37475825]
[-0.11157634  0.21503249]
[-1.89415691  0.44697764]
[ 1.67100423  1.75166912]
[-0.30964085 -1.37959044]
[-0.30964085 -0.65476184]
[ 0.8787462   2.15757314]
[ 0.28455268 -0.53878926]
[ 0.8787462   1.02684052]
[-1.49802789 -1.20563157]
[ 1.07681071  2.07059371]
[-1.00286662  0.50496393]
[-0.90383437  0.30201192]
[-0.11157634 -0.21986468]
[-0.60673761  0.47597078]
[-1.6960924   0.53395707]
[-0.11157634  0.27301877]
[ 1.86906873 -0.27785096]
[-0.11157634 -0.48080297]
[-1.39899564 -0.33583725]
[-1.99318916 -0.50979612]
[-1.59706014  0.33100506]
[-0.4086731  -0.77073441]
[-0.70576986 -1.03167271]
[ 1.07681071 -0.97368642]
[-1.10189888  0.53395707]
[ 0.28455268 -0.50979612]
[-1.10189888  0.41798449]
[-0.30964085 -1.43757673]
[ 0.48261718  1.22979253]
[-1.10189888 -0.33583725]
[-0.11157634  0.30201192]
[ 1.37390747  0.59194336]
[-1.20093113 -1.14764529]
[ 1.07681071  0.47597078]
[ 1.86906873  1.51972397]
[-0.4086731  -1.29261101]
[-0.30964085 -0.3648304 ]
[-0.4086731   1.31677196]
[ 2.06713324  0.53395707]
[ 0.68068169 -1.089659  ]
[-0.90383437  0.38899135]
[-1.20093113  0.30201192]
```

```
 [ 1.07681071 -1.20563157]
 [-1.49802789 -1.43757673]
 [-0.60673761 -1.49556302]
 [ 2.1661655  -0.79972756]
 [-1.89415691  0.18603934]
 [-0.21060859  0.85288166]
 [-1.89415691 -1.26361786]
 [ 2.1661655   0.38899135]
 [-1.39899564  0.56295021]
 [-1.10189888 -0.33583725]
 [ 0.18552042 -0.65476184]
 [ 0.38358493  0.01208048]
 [-0.60673761  2.331532  ]
 [-0.30964085  0.21503249]
 [-1.59706014 -0.19087153]
 [ 0.68068169 -1.37959044]
 [-1.10189888  0.56295021]
 [-1.99318916  0.35999821]
 [ 0.38358493  0.27301877]
 [ 0.18552042 -0.27785096]
 [ 1.47293972 -1.03167271]
 [ 0.8787462   1.08482681]
 [ 1.96810099  2.15757314]
 [ 2.06713324  0.38899135]
 [-1.39899564 -0.42281668]
 [-1.20093113 -1.00267957]
 [ 1.96810099 -0.91570013]
 [ 0.38358493  0.30201192]
 [ 0.18552042  0.1570462 ]
 [ 2.06713324  1.75166912]
 [ 0.77971394 -0.8287207 ]
 [ 0.28455268 -0.27785096]
 [ 0.38358493 -0.16187839]
 [-0.11157634  2.21555943]
 [-1.49802789 -0.62576869]
 [-1.29996338 -1.06066585]
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 [-1.10189888  0.76590222]
 [-1.49802789 -0.19087153]
 [ 0.97777845 -1.06066585]
 [ 0.97777845  0.59194336]
 [ 0.38358493  0.99784738]]
```

Training the K-NN model on the Training set

```
from sklearn.neighbors import KNeighborsClassifier
classifier = KNeighborsClassifier(n_neighbors = 5, metric = 'minkowski', p = 2)
classifier.fit(X_train, y_train)
```

Predicting a new result

```
print(classifier.predict(sc.transform([[30,87000]])))
[0]
```

Predicting the Test set results

```
y_pred = classifier.predict(X_test)
print(np.concatenate((y_pred.reshape(len(y_pred),1), y_test.reshape(len(y_test),1)),1))

[[0 0]
 [0 0]
 [0 0]
 [0 0]
 [0 0]
 [0 0]
 [0 0]
 [1 1]
 [0 0]
 [1 0]
 [0 0]
 [0 0]
 [0 0]
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 [1 0]
 [0 0]
 [0 0]
 [1 1]
 [0 0]
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 [1 1]
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 [0 0]
 [0 0]
 [0 0]
 [0 0]
 [0 1]
 [1 1]
 [0 0]
 [0 0]
 [0 0]
 [0 0]
```

[0 0]
[0 0]
[1 1]
[0 0]
[0 0]
[0 0]
[0 0]
[1 1]
[0 0]
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[1 1]
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[0 0]

```
 [1 1]
 [1 1]
 [0 0]
 [0 0]
 [1 1]
 [0 0]
 [0 0]
 [0 0]
 [0 1]
 [0 0]
 [1 1]
 [1 1]
 [1 1]]
```

Making the Confusion Matrix

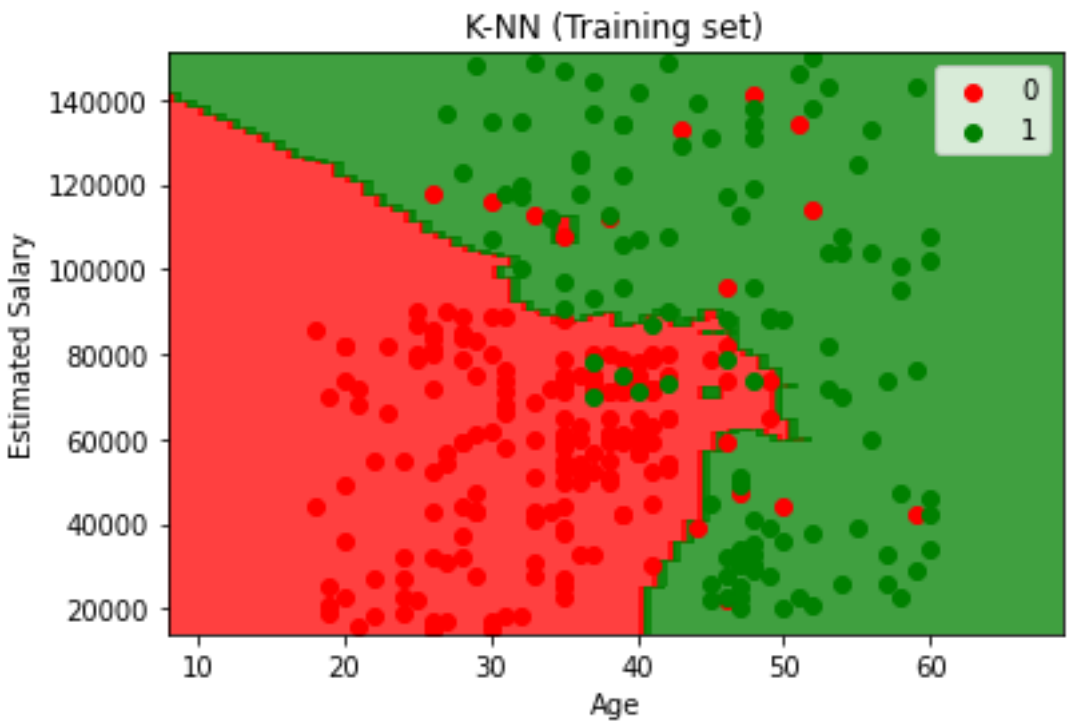
```
from sklearn.metrics import confusion_matrix, accuracy_score
cm = confusion_matrix(y_test, y_pred)
print(cm)
accuracy_score(y_test, y_pred)
```

```
[[64  4]
 [ 3 29]]
```

```
0.93
```

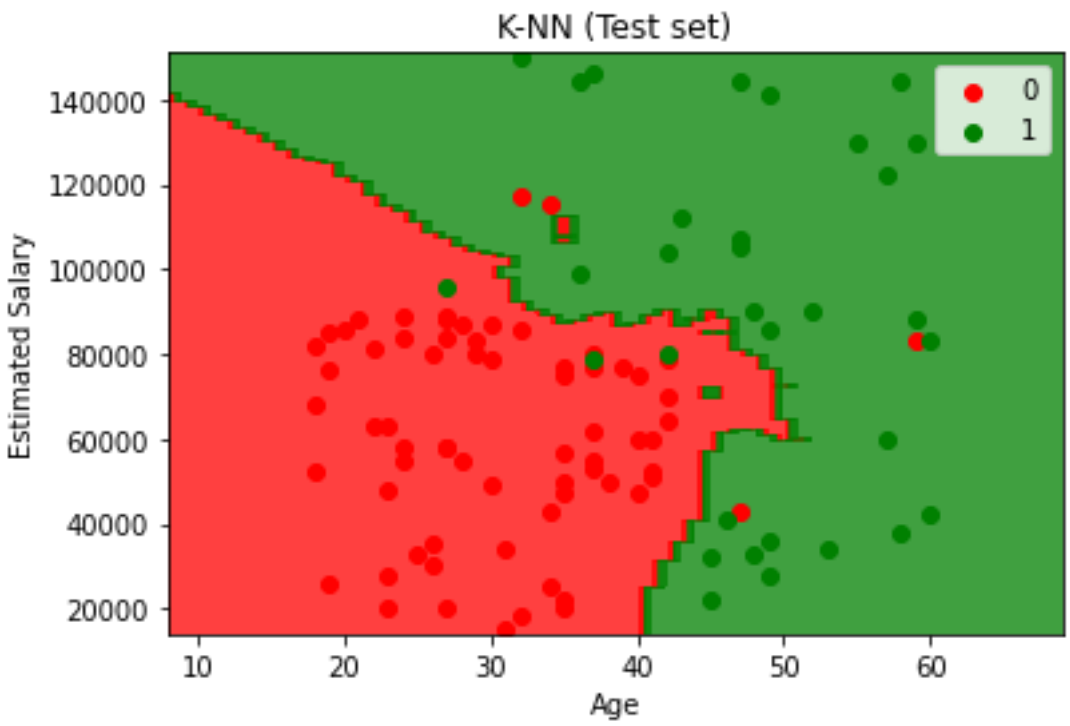
Visualising the Training set results

```
from matplotlib.colors import ListedColormap
X_set, y_set = sc.inverse_transform(X_train), y_train
X1, X2 = np.meshgrid(np.arange(start = X_set[:, 0].min() - 10, stop = X_set[:, 0].max() + 10, step =
1),
 np.arange(start = X_set[:, 1].min() - 1000, stop = X_set[:, 1].max() + 1000, step = 1))
plt.contourf(X1, X2, classifier.predict(sc.transform(np.array([X1.ravel(),
X2.ravel()]).T)).reshape(X1.shape),
             alpha = 0.75, cmap = ListedColormap(('red', 'green')))
plt.xlim(X1.min(), X1.max())
plt.ylim(X2.min(), X2.max())
for i, j in enumerate(np.unique(y_set)):
    plt.scatter(X_set[y_set == j, 0], X_set[y_set == j, 1], c = ListedColormap(('red', 'green'))(i), label
= j)
plt.title('K-NN (Training set)')
plt.xlabel('Age')
plt.ylabel('Estimated Salary')
plt.legend()
plt.show()
```



Visualising the Test set results

```
from matplotlib.colors import ListedColormap
X_set, y_set = sc.inverse_transform(X_test), y_test
X1, X2 = np.meshgrid(np.arange(start = X_set[:, 0].min() - 10, stop = X_set[:, 0].max() + 10, step =
1),
np.arange(start = X_set[:, 1].min() - 1000, stop = X_set[:, 1].max() + 1000, step = 1))
plt.contourf(X1, X2, classifier.predict(sc.transform(np.array([X1.ravel(),
X2.ravel()]).T)).reshape(X1.shape),
             alpha = 0.75, cmap = ListedColormap(('red', 'green')))
plt.xlim(X1.min(), X1.max())
plt.ylim(X2.min(), X2.max())
for i, j in enumerate(np.unique(y_set)):
    plt.scatter(X_set[y_set == j, 0], X_set[y_set == j, 1], c = ListedColormap(('red', 'green'))(i), label
= j)
plt.title('K-NN (Test set)')
plt.xlabel('Age')
plt.ylabel('Estimated Salary')
plt.legend()
plt.show()
```



Support Vector Machine (SVM)

Importing the libraries

```
import numpy as np
import matplotlib.pyplot as plt
import pandas as pd
```

Importing the dataset

```
dataset = pd.read_csv('Social_Network_Ads.csv')
X = dataset.iloc[:, :-1].values
y = dataset.iloc[:, -1].values
```

Splitting the dataset into the Training set and Test set

```
from sklearn.model_selection import train_test_split
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size = 0.25, random_state = 0)
```

print(X_train)

```
[[    44  39000]
```

```
[    32 120000]
[    38  50000]
[    32 135000]
[    52  21000]
[    53 104000]
[    39  42000]
[    38  61000]
[    36  50000]
[    36  63000]
[    35  25000]
[    35  50000]
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```

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```

```
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[    40  59000]
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[    39  71000]
[    20  74000]
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[    31  76000]
[    47  25000]
```

```
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```

```
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[    26  72000]
[    35  23000]
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[    30  17000]
[    39 134000]
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[    33  43000]
```

```
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[    26  32000]
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[    39  73000]
[    41  79000]
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[    41  30000]
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[    42 108000]
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[    34  43000]
[    37  52000]
[    48  30000]
[    29  43000]
[    36  52000]
[    27  54000]
[    26 118000]]
```

print(y_train)

```
[0 1 0 1 1 1 0 0 0 0 0 0 1 1 1 0 1 0 0 1 0 1 0 1 0 0 1 1 1 1 0 1 0 1 0 0 1
 0 0 1 0 0 0 0 0 1 1 1 1 0 0 0 1 0 1 0 1 0 0 1 0 0 0 1 0 0 0 1 1 0 0 1 0 1
 1 1 0 0 1 1 0 0 1 1 0 1 0 0 1 1 0 1 1 1 0 0 0 0 0 1 0 0 1 1 1 1 1 0 1 1 0
 1 0 0 0 0 0 0 0 1 1 0 0 1 0 0 1 0 0 0 1 0 1 1 0 1 0 0 0 0 1 0 0 0 1 1 0 0
 0 0 1 0 1 0 0 0 1 0 0 0 0 1 1 1 0 0 0 0 0 0 1 1 1 1 1 0 1 0 0 0 0 0 1 0 0
 0 0 0 0 1 1 0 1 0 1 0 0 1 0 0 0 1 0 0 0 0 0 1 0 0 0 0 0 1 0 1 1 0 0 0 0 0
 0 1 1 0 0 0 0 1 0 0 0 0 1 0 1 0 1 0 0 0 1 0 0 0 1 0 1 0 0 0 0 0 1 1 0 0 0
 0 0 1 0 1 1 0 0 0 0 0 1 0 1 0 0 1 0 0 1 0 1 0 0 0 0 0 0 1 1 1 1 0 0 0 0 1
 0 0 0 0]
```

print(X_test)

```
[[    30  87000]
 [    38  50000]
 [    35  75000]
 [    30  79000]
 [    35  50000]
 [    27  20000]
 [    31  15000]
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 [    47  43000]
 [    30  49000]
 [    28  55000]
 [    37  55000]
 [    39  77000]
 [    20  86000]
 [    32 117000]
 [    37  77000]
 [    19  85000]
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 [    47 105000]
 [    23  28000]
 [    49 141000]
 [    28  87000]
 [    29  80000]
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 [    32  86000]
 [    21  88000]
 [    37  79000]
 [    57  60000]
 [    37  53000]
 [    24  58000]
 [    18  52000]
```

```
[    22  81000]
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[    31  34000]
[    49  36000]
[    27  88000]
[    41  52000]
[    27  84000]
[    35  20000]
[    43 112000]
[    27  58000]
[    37  80000]
[    52  90000]
[    26  30000]
[    49  86000]
[    57 122000]
[    34  25000]
[    35  57000]
[    34 115000]
[    59  88000]
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[    23  20000]
[    32  18000]
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[    40  47000]
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[    32 150000]
[    35  77000]
[    22  63000]
[    45  22000]
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[    42  79000]
[    40  60000]
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[    47 107000]
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[    59  83000]
[    24  55000]
[    26  35000]
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[    42  80000]
```

```
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 [    59 130000]
 [    46  41000]
 [    41  60000]
 [    42  64000]
 [    37 146000]
 [    23  48000]
 [    25  33000]
 [    24  84000]
 [    27  96000]
 [    23  63000]
 [    48  33000]
 [    48  90000]
 [    42 104000]]
```

print(y_test)

```
[0 0 0 0 0 0 0 1 0 0 0 0 0 0 0 0 0 0 1 0 0 1 0 1 0 1 0 0 0 0 0 1 1 0 0 0 0
 0 0 1 0 0 0 0 1 0 0 1 0 1 1 0 0 0 1 1 0 0 1 0 0 1 0 1 0 1 0 0 0 0 1 0 0 1
 0 0 0 0 1 1 1 0 0 0 1 1 0 1 1 0 0 1 0 0 0 1 0 1 1 1]
```

Feature Scaling

```
from sklearn.preprocessing import StandardScaler
sc = StandardScaler()
X_train = sc.fit_transform(X_train)
X_test = sc.transform(X_test)
```

print(X_train)

```
[[ 0.58164944 -0.88670699]
 [-0.60673761  1.46173768]
 [-0.01254409 -0.5677824 ]
 [-0.60673761  1.89663484]
 [ 1.37390747 -1.40858358]
 [ 1.47293972  0.99784738]
 [ 0.08648817 -0.79972756]
 [-0.01254409 -0.24885782]
 [-0.21060859 -0.5677824 ]
 [-0.21060859 -0.19087153]
 [-0.30964085 -1.29261101]
 [-0.30964085 -0.5677824 ]
 [ 0.38358493  0.09905991]
 [ 0.8787462  -0.59677555]
 [ 2.06713324 -1.17663843]
 [ 1.07681071 -0.13288524]
 [ 0.68068169  1.78066227]
 [-0.70576986  0.56295021]
 [ 0.77971394  0.35999821]
```

```
[ 0.8787462  -0.53878926]
[-1.20093113 -1.58254245]
[ 2.1661655   0.93986109]
[-0.01254409  1.22979253]
[ 0.18552042  1.08482681]
[ 0.38358493 -0.48080297]
[-0.30964085 -0.30684411]
[ 0.97777845 -0.8287207 ]
[ 0.97777845  1.8676417 ]
[-0.01254409  1.25878567]
[-0.90383437  2.27354572]
[-1.20093113 -1.58254245]
[ 2.1661655  -0.79972756]
[-1.39899564 -1.46656987]
[ 0.38358493  2.30253886]
[ 0.77971394  0.76590222]
[-1.00286662 -0.30684411]
[ 0.08648817  0.76590222]
[-1.00286662  0.56295021]
[ 0.28455268  0.07006676]
[ 0.68068169 -1.26361786]
[-0.50770535 -0.01691267]
[-1.79512465  0.35999821]
[-0.70576986  0.12805305]
[ 0.38358493  0.30201192]
[-0.30964085  0.07006676]
[-0.50770535  2.30253886]
[ 0.18552042  0.04107362]
[ 1.27487521  2.21555943]
[ 0.77971394  0.27301877]
[-0.30964085  0.1570462 ]
[-0.01254409 -0.53878926]
[-0.21060859  0.1570462 ]
[-0.11157634  0.24402563]
[-0.01254409 -0.24885782]
[ 2.1661655   1.11381995]
[-1.79512465  0.35999821]
[ 1.86906873  0.12805305]
[ 0.38358493 -0.13288524]
[-1.20093113  0.30201192]
[ 0.77971394  1.37475825]
[-0.30964085 -0.24885782]
[-1.6960924  -0.04590581]
[-1.00286662 -0.74174127]
[ 0.28455268  0.50496393]
[-0.11157634 -1.06066585]
[-1.10189888  0.59194336]
[ 0.08648817 -0.79972756]
[-1.00286662  1.54871711]
[-0.70576986  1.40375139]
```

```
[-1.29996338  0.50496393]
[-0.30964085  0.04107362]
[-0.11157634  0.01208048]
[-0.30964085 -0.88670699]
[ 0.8787462  -1.3505973 ]
[-0.30964085  2.24455257]
[ 0.97777845  1.98361427]
[-1.20093113  0.47597078]
[-1.29996338  0.27301877]
[ 1.37390747  1.98361427]
[ 1.27487521 -1.3505973 ]
[-0.30964085 -0.27785096]
[-0.50770535  1.25878567]
[-0.80480212  1.08482681]
[ 0.97777845 -1.06066585]
[ 0.28455268  0.30201192]
[ 0.97777845  0.76590222]
[-0.70576986 -1.49556302]
[-0.70576986  0.04107362]
[ 0.48261718  1.72267598]
[ 2.06713324  0.18603934]
[-1.99318916 -0.74174127]
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[ 0.38358493  0.59194336]
[ 0.8787462  -1.14764529]
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[ 0.77971394 -0.30684411]
[ 2.06713324 -0.79972756]
[ 0.77971394  0.12805305]
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[ 1.86906873 -1.26361786]
[ 1.37390747 -0.91570013]
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[ 1.47293972  2.12857999]
[-0.30964085 -1.23462472]
[ 1.96810099  0.91086794]
[ 0.68068169 -0.71274813]
[-1.49802789  0.35999821]
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[-0.01254409 -0.30684411]
[-1.20093113  0.41798449]
[-0.90383437 -1.20563157]
[-0.11157634  0.04107362]
[-1.59706014 -0.42281668]
```

```
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[-1.10189888 -1.52455616]
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[ 0.08648817 -0.30684411]
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[-0.60673761 -1.49556302]
[ 0.77971394  0.53395707]
[-0.30964085 -0.33583725]
[ 1.77003648 -0.27785096]
[ 0.8787462  -1.03167271]
[ 0.18552042  0.07006676]
[-0.60673761  0.8818748 ]
[-1.89415691 -1.40858358]
[-1.29996338  0.59194336]
[-0.30964085  0.53395707]
[-1.00286662 -1.089659  ]
[ 1.17584296 -1.43757673]
[ 0.18552042 -0.30684411]
[ 1.17584296 -0.74174127]
[-0.30964085  0.07006676]
[ 0.18552042  2.09958685]
[ 0.77971394 -1.089659  ]
[ 0.08648817  0.04107362]
[-1.79512465  0.12805305]
[-0.90383437  0.1570462 ]
[-0.70576986  0.18603934]
[ 0.8787462  -1.29261101]
[ 0.18552042 -0.24885782]
[-0.4086731   1.22979253]
[-0.01254409  0.30201192]
[ 0.38358493  0.1570462 ]
[ 0.8787462  -0.65476184]
[ 0.08648817  0.1570462 ]
[-1.89415691 -1.29261101]
[-0.11157634  0.30201192]
[-0.21060859 -0.27785096]
[ 0.28455268 -0.50979612]
[-0.21060859  1.6067034 ]
[ 0.97777845 -1.17663843]
[-0.21060859  1.63569655]
[ 1.27487521  1.8676417 ]
[-1.10189888 -0.3648304 ]
[-0.01254409  0.04107362]
[ 0.08648817 -0.24885782]
[-1.59706014 -1.23462472]
```

```
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[ 1.47293972  0.07006676]
[-0.60673761  1.37475825]
[ 1.57197197  0.01208048]
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[ 0.68068169  0.27301877]
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[-1.89415691 -1.46656987]
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[ 0.08648817  0.04107362]
[-1.39899564 -1.3505973 ]
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[-1.20093113 -1.5535493 ]
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[-1.00286662  0.27301877]
[ 1.47293972  0.35999821]
[ 0.18552042 -0.3648304 ]
[ 2.1661655  -1.03167271]
[-0.30964085  1.11381995]
[-1.6960924   0.07006676]
[-0.01254409  0.04107362]
[ 0.08648817  1.05583366]
[-0.11157634 -0.3648304 ]
[-1.20093113  0.07006676]
[-0.30964085 -1.3505973 ]
[ 1.57197197  1.11381995]
[-0.80480212 -1.52455616]
[ 0.08648817  1.8676417 ]
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[ 0.28455268 -0.71274813]
[ 0.28455268  0.07006676]
[ 0.08648817  1.8676417 ]
[-1.10189888  1.95462113]
[-1.6960924  -1.5535493 ]
[-1.20093113 -1.089659  ]
[-0.70576986 -0.1038921 ]
[ 0.08648817  0.09905991]
[ 0.28455268  0.27301877]
[ 0.8787462  -0.5677824 ]
[ 0.28455268 -1.14764529]
[-0.11157634  0.67892279]
[ 2.1661655  -0.68375498]
[-1.29996338 -1.37959044]
[-1.00286662 -0.94469328]
[-0.01254409 -0.42281668]
[-0.21060859 -0.45180983]
```

```
 [-1.79512465 -0.97368642]
 [ 1.77003648  0.99784738]
 [ 0.18552042 -0.3648304 ]
 [ 0.38358493  1.11381995]
 [-1.79512465 -1.3505973 ]
 [ 0.18552042 -0.13288524]
 [ 0.8787462  -1.43757673]
 [-1.99318916  0.47597078]
 [-0.30964085  0.27301877]
 [ 1.86906873 -1.06066585]
 [-0.4086731   0.07006676]
 [ 1.07681071 -0.88670699]
 [-1.10189888 -1.11865214]
 [-1.89415691  0.01208048]
 [ 0.08648817  0.27301877]
 [-1.20093113  0.33100506]
 [-1.29996338  0.30201192]
 [-1.00286662  0.44697764]
 [ 1.67100423 -0.88670699]
 [ 1.17584296  0.53395707]
 [ 1.07681071  0.53395707]
 [ 1.37390747  2.331532  ]
 [-0.30964085 -0.13288524]
 [ 0.38358493 -0.45180983]
 [-0.4086731  -0.77073441]
 [-0.11157634 -0.50979612]
 [ 0.97777845 -1.14764529]
 [-0.90383437 -0.77073441]
 [-0.21060859 -0.50979612]
 [-1.10189888 -0.45180983]
 [-1.20093113  1.40375139]]
```

print(X_test)

```
[[-0.80480212  0.50496393]
 [-0.01254409 -0.5677824 ]
 [-0.30964085  0.1570462 ]
 [-0.80480212  0.27301877]
 [-0.30964085 -0.5677824 ]
 [-1.10189888 -1.43757673]
 [-0.70576986 -1.58254245]
 [-0.21060859  2.15757314]
 [-1.99318916 -0.04590581]
 [ 0.8787462  -0.77073441]
 [-0.80480212 -0.59677555]
 [-1.00286662 -0.42281668]
 [-0.11157634 -0.42281668]
 [ 0.08648817  0.21503249]
 [-1.79512465  0.47597078]
 [-0.60673761  1.37475825]
```

```
[-0.11157634  0.21503249]
[-1.89415691  0.44697764]
[ 1.67100423  1.75166912]
[-0.30964085 -1.37959044]
[-0.30964085 -0.65476184]
[ 0.8787462   2.15757314]
[ 0.28455268 -0.53878926]
[ 0.8787462   1.02684052]
[-1.49802789 -1.20563157]
[ 1.07681071  2.07059371]
[-1.00286662  0.50496393]
[-0.90383437  0.30201192]
[-0.11157634 -0.21986468]
[-0.60673761  0.47597078]
[-1.6960924   0.53395707]
[-0.11157634  0.27301877]
[ 1.86906873 -0.27785096]
[-0.11157634 -0.48080297]
[-1.39899564 -0.33583725]
[-1.99318916 -0.50979612]
[-1.59706014  0.33100506]
[-0.4086731  -0.77073441]
[-0.70576986 -1.03167271]
[ 1.07681071 -0.97368642]
[-1.10189888  0.53395707]
[ 0.28455268 -0.50979612]
[-1.10189888  0.41798449]
[-0.30964085 -1.43757673]
[ 0.48261718  1.22979253]
[-1.10189888 -0.33583725]
[-0.11157634  0.30201192]
[ 1.37390747  0.59194336]
[-1.20093113 -1.14764529]
[ 1.07681071  0.47597078]
[ 1.86906873  1.51972397]
[-0.4086731  -1.29261101]
[-0.30964085 -0.3648304 ]
[-0.4086731   1.31677196]
[ 2.06713324  0.53395707]
[ 0.68068169 -1.089659  ]
[-0.90383437  0.38899135]
[-1.20093113  0.30201192]
[ 1.07681071 -1.20563157]
[-1.49802789 -1.43757673]
[-0.60673761 -1.49556302]
[ 2.1661655  -0.79972756]
[-1.89415691  0.18603934]
[-0.21060859  0.85288166]
[-1.89415691 -1.26361786]
[ 2.1661655   0.38899135]
```

```
 [-1.39899564  0.56295021]
 [-1.10189888 -0.33583725]
 [ 0.18552042 -0.65476184]
 [ 0.38358493  0.01208048]
 [-0.60673761  2.331532  ]
 [-0.30964085  0.21503249]
 [-1.59706014 -0.19087153]
 [ 0.68068169 -1.37959044]
 [-1.10189888  0.56295021]
 [-1.99318916  0.35999821]
 [ 0.38358493  0.27301877]
 [ 0.18552042 -0.27785096]
 [ 1.47293972 -1.03167271]
 [ 0.8787462   1.08482681]
 [ 1.96810099  2.15757314]
 [ 2.06713324  0.38899135]
 [-1.39899564 -0.42281668]
 [-1.20093113 -1.00267957]
 [ 1.96810099 -0.91570013]
 [ 0.38358493  0.30201192]
 [ 0.18552042  0.1570462 ]
 [ 2.06713324  1.75166912]
 [ 0.77971394 -0.8287207 ]
 [ 0.28455268 -0.27785096]
 [ 0.38358493 -0.16187839]
 [-0.11157634  2.21555943]
 [-1.49802789 -0.62576869]
 [-1.29996338 -1.06066585]
 [-1.39899564  0.41798449]
 [-1.10189888  0.76590222]
 [-1.49802789 -0.19087153]
 [ 0.97777845 -1.06066585]
 [ 0.97777845  0.59194336]
 [ 0.38358493  0.99784738]]
```

Training the SVM model on the Training set

```
from sklearn.svm import SVC
classifier = SVC(kernel = 'linear', random_state = 0)
classifier.fit(X_train, y_train)
```

Predicting a new result

```
print(classifier.predict(sc.transform([[30,87000]])))
```

```
[0]
```

Predicting the Test set results

```
y_pred = classifier.predict(X_test)

print(np.concatenate((y_pred.reshape(len(y_pred),1), y_test.reshape(len(y_test),1)),1))
```

```
[[0 0]
 [0 0]
 [0 0]
 [0 0]
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 [0 0]
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 [0 0]
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 [0 0]
 [0 1]
 [0 0]
 [0 1]
 [1 1]
 [1 1]]
```

Making the Confusion Matrix

```
from sklearn.metrics import confusion_matrix, accuracy_score

cm = confusion_matrix(y_test, y_pred)
```

print(cm)

```
accuracy_score(y_test, y_pred)
```

```
[[66  2]
 [ 8 24]]
```

```
0.9
```

Visualising the Training set results

```
from matplotlib.colors import ListedColormap

X_set, y_set = sc.inverse_transform(X_train), y_train

X1, X2 = np.meshgrid(np.arange(start = X_set[:, 0].min() - 10, stop = X_set[:, 0].max() + 10, step =
0.25),

np.arange(start = X_set[:, 1].min() - 1000, stop = X_set[:, 1].max() + 1000, step = 0.25))

plt.contourf(X1, X2, classifier.predict(sc.transform(np.array([X1.ravel(),
X2.ravel()]).T)).reshape(X1.shape),

            alpha = 0.75, cmap = ListedColormap(('red', 'green')))

plt.xlim(X1.min(), X1.max())

plt.ylim(X2.min(), X2.max())

for i, j in enumerate(np.unique(y_set)):

    plt.scatter(X_set[y_set == j, 0], X_set[y_set == j, 1], c = ListedColormap(('red', 'green'))(i), label
= j)

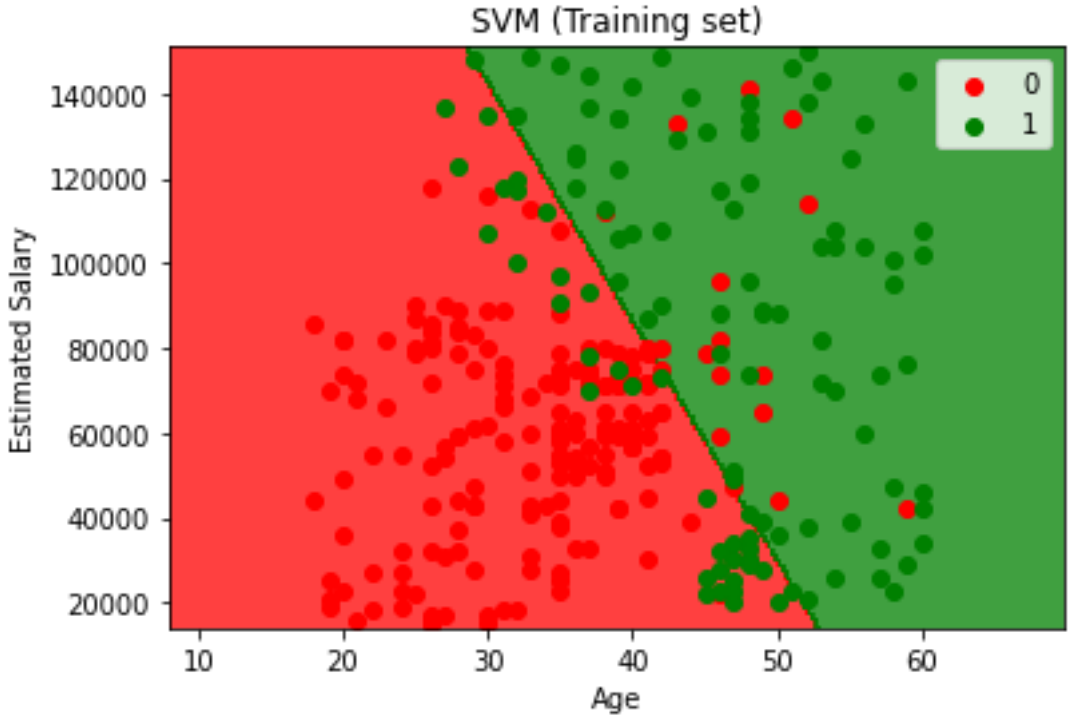
plt.title('SVM (Training set)')
```

```
plt.xlabel('Age')

plt.ylabel('Estimated Salary')

plt.legend()

plt.show()
```



Visualising the Test set results

```
from matplotlib.colors import ListedColormap

X_set, y_set = sc.inverse_transform(X_test), y_test

X1, X2 = np.meshgrid(np.arange(start = X_set[:, 0].min() - 10, stop = X_set[:, 0].max() + 10, step =
0.25),

np.arange(start = X_set[:, 1].min() - 1000, stop = X_set[:, 1].max() + 1000, step = 0.25))

plt.contourf(X1, X2, classifier.predict(sc.transform(np.array([X1.ravel(),
X2.ravel()]).T)).reshape(X1.shape),

            alpha = 0.75, cmap = ListedColormap(('red', 'green')))

plt.xlim(X1.min(), X1.max())

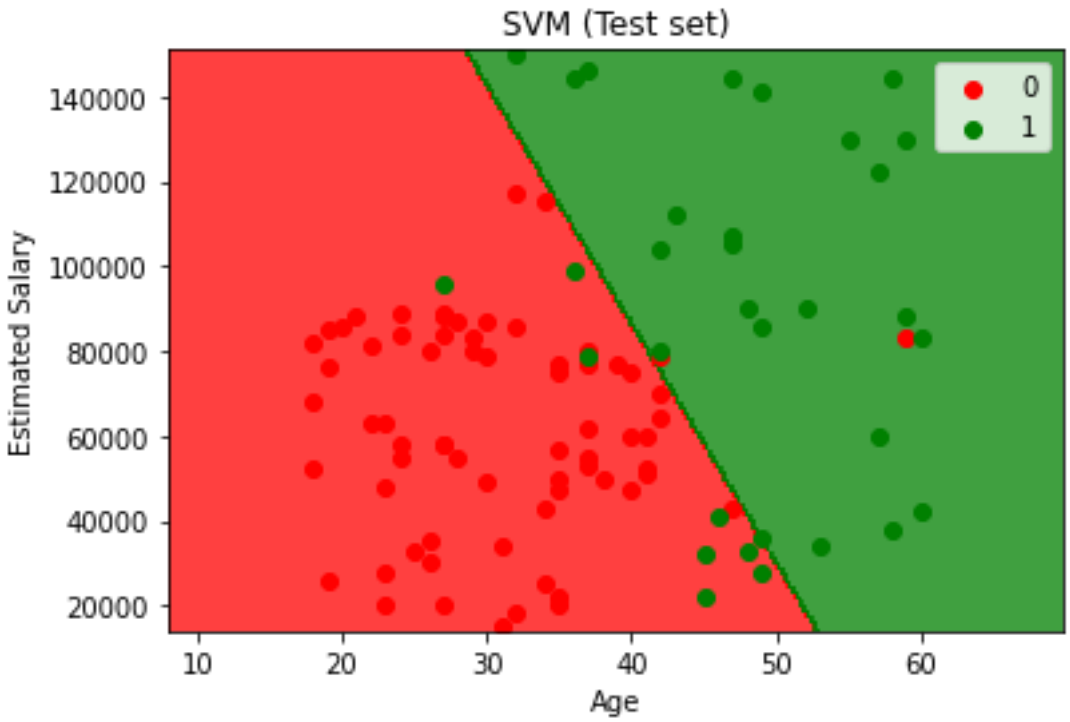
plt.ylim(X2.min(), X2.max())

for i, j in enumerate(np.unique(y_set)):

    plt.scatter(X_set[y_set == j, 0], X_set[y_set == j, 1], c = ListedColormap(('red', 'green'))(i), label
= j)

plt.title('SVM (Test set)')
```

```
plt.xlabel('Age')
plt.ylabel('Estimated Salary')
plt.legend()
plt.show()
```



***